

Vanessa Kacaj
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Education

Cornell University, College of Engineering

Major: Mechanical Engineering

Ithaca, NY

Anticipated Graduation Date: May 2028

Relevant Coursework:

- Engineering Applications of Operations Research, Engineering General Chemistry, Calculus for Engineers, Multivariable Calculus for Engineers, Physics I: Mechanics and Heat, Introduction to Experimental Physics, Statics and Mechanics of Solids, Thermodynamics, Differential Equations for Engineers, Physics II: Electromagnetism

John F. Kennedy High School

New York State Advanced Regents Diploma 2024

Bellmore, NY

Graduation Date: May 2024

Extracurriculars

Member of the John F. Kennedy Robotics club

(2020- 2024)

- Vex Robotics Competition: Two-time Judges Award Recipient
- Vex Robotics New York State Championships: 2023 and 2024
- Met frequently with team to brainstorm, design, and build our robot
- Responsible for maintaining detailed engineering notebook to document design process.

Community, Health, and Research in Medicine Committee

(2021-2024)

- Raised money for the Cerebral Palsy Foundation through fun events such as a trivia challenge and medical Scrabble
- Taught children with cerebral palsy about robotics by running small workshops so that they could test out and drive robots

Member of Cornell Concrete Canoe

(Sept. 2025- Present)

- Every fall, develop a brand-new canoe engineered from concrete mixtures to compete in the spring ASCE Regional Concrete Canoe Competition.
- Member of the mold sub-team, responsible for construction and design of the wooden mold, and subsequently, the design of the concrete canoe, casting the concrete canoe, and writing the construction section of the design report

Simons Electron Microscopy Center Cryo-EM course

(2022)

Experience

Advanced Science Research Program *Bellmore, NY*

(Sept. 2021- Jun. 2024)

- Conduct an average of 15 hours of biweekly independent research on RNA binding protein Msi2 by reading scientific journal articles and exploring websites

Memorial Sloan Kettering Cancer Center Dinshaw Patel Laboratory *New York, NY*

(Jul. 2022- Nov. 2024)

- Conducted independent graduate level research on RNA binding protein Msi2, wrote a research manuscript, and assisted in various different research groups
- Wanted to determine the three-dimensional structure of Msi2 and also what RNA sequences it most likely binds to in order to be able to create cancer therapeutics for this protein

Liberty Maritime Corporation Engineering Intern *New Hyde Park, NY*

(Jun. 2025-Jul.2025)

- Reviewed deck drawings for a PCTC cargo ship to support a deck strengthening project aimed at enhancing the vessel's utility, working alongside naval designers to meet structural requirements set by the U.S. military
- Researched alternative energy sources for PCTC and bulkers to reduce fuel use and emissions; collaborated with naval architects on integration strategies.

Awards:

- High Honor Roll
- Language: Seal of Biliteracy in Spanish
- Health Career Development Certificate

Skills:

Micro-pipetting | X-Ray Crystallography | PCR | Genotyping | Cryo-EM | Gel Electrophoresis | Microsoft Suite | Naval Architecture Basics | Structural Analysis & Design Understanding | Nautical Systems